

Table 1: Average long-term forecasting results on the benchmark datasets. We followed the same setting in DMMV [Shen et al., 2025]. We set the forecasting horizons  $H \in \{24, 36, 48, 60\}$  for Illness and  $\{96, 192, 336, 720\}$  for the others. **Red**: the best, Blue: the second best.

| View        | Language             |              |        |              |          |       | Multi-Modal  |              |              |              | Visual       |              | Numerical    |              |          |       |
|-------------|----------------------|--------------|--------|--------------|----------|-------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|----------|-------|
| Methods     | RDTU ( <b>Ours</b> ) |              | GPT4TS |              | Time-LLM |       | DMMV-A       |              | Time-VLM     |              | VisionTS     |              | PatchTST     |              | TimesNet |       |
| Metric      | MSE                  | MAE          | MSE    | MAE          | MSE      | MAE   | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          | MSE      | MAE   |
| ETTh1       | <u>0.399</u>         | <b>0.413</b> | 0.418  | 0.421        | 0.418    | 0.432 | <b>0.395</b> | <u>0.414</u> | 0.405        | 0.420        | 0.407        | 0.419        | 0.413        | 0.431        | 0.458    | 0.450 |
| ETTh2       | <u>0.337</u>         | <b>0.377</b> | 0.354  | 0.389        | 0.361    | 0.396 | <u>0.337</u> | 0.388        | 0.341        | 0.391        | 0.351        | 0.386        | <b>0.330</b> | <u>0.379</u> | 0.414    | 0.427 |
| ETTm1       | <b>0.340</b>         | <b>0.367</b> | 0.363  | 0.378        | 0.356    | 0.377 | <b>0.340</b> | <u>0.371</u> | 0.351        | 0.376        | <u>0.344</u> | 0.373        | 0.351        | 0.381        | 0.400    | 0.406 |
| ETTm2       | <u>0.250</u>         | <b>0.309</b> | 0.254  | <u>0.311</u> | 0.261    | 0.316 | 0.256        | 0.317        | <b>0.248</b> | <u>0.311</u> | 0.267        | 0.327        | 0.255        | 0.315        | 0.291    | 0.333 |
| Illness     | 1.559                | 0.812        | 1.871  | 0.852        | 2.018    | 0.894 | <b>1.407</b> | <b>0.771</b> | –            | –            | 1.482        | <u>0.796</u> | <u>1.443</u> | 0.798        | 2.139    | 0.931 |
| Electricity | <b>0.156</b>         | <b>0.246</b> | 0.170  | 0.263        | 0.165    | 0.259 | <u>0.158</u> | <u>0.248</u> | 0.172        | 0.272        | 0.159        | 0.250        | 0.162        | 0.253        | 0.193    | 0.295 |
| Weather     | <u>0.221</u>         | <b>0.251</b> | 0.227  | <u>0.255</u> | 0.244    | 0.270 | <b>0.217</b> | 0.256        | 0.224        | 0.263        | 0.225        | 0.258        | 0.226        | 0.264        | 0.259    | 0.287 |
| Traffic     | <u>0.389</u>         | <b>0.255</b> | 0.421  | 0.274        | 0.422    | 0.281 | <u>0.389</u> | 0.257        | 0.419        | 0.304        | <b>0.386</b> | <u>0.256</u> | 0.391        | 0.264        | 0.620    | 0.336 |
| # Wins      | <b>9</b>             |              | 0      |              | 0        |       | <u>5</u>     |              | 1            |              | 1            |              | 1            |              | 0        |       |